

EXERCISE - 2.1

1. If $a + b + c = 66$ and $ab + bc + ca = 1071$, where a, b and c are all prime numbers, find the value of abc .
 (1) 1007 (2) 2876 (3) 1886 (4) 1765
2. The positive integers $A, B, A - B$ and $A + B$ are all prime numbers. The sum of these four primes is
 (1) Even (2) Divisible by 3 (3) Prime (4) Divisible by 7
3. For how many positive integers a is $a^4 - 3a^2 + 9$ a prime number ?
 (1) 5 (2) 7 (3) 6 (4) 2
4. If n is a positive integer greater than 1, then $n^4 + 4$ is a _____ number.
 (1) Composite (2) Composite odd number
 (3) Composite even number (4) Prime
5. What is the smallest positive integer x for which $x^2 - x + 5$ is not a prime ?
 (1) 5 (2) 7 (3) 6 (4) 13
6. What is the smallest composite number generated by $p^2 - p - 1$ where p is a prime ?
 (1) 155 (2) 157 (3) 156 (4) 154
7. Some prime numbers are 1 more than a square, as in $4^2 + 1 = 17$. How many primes $p \leq 100$ are of the form $p = n^2 + 1$?
 (1) 5 (2) 7 (3) 4 (4) 23
8. How many prime numbers satisfy $\sqrt{300} < x < \sqrt{700}$?
 (1) 5 numbers (2) 3 numbers (3) 2 numbers (4) 1 number
9. The number of pairs of prime numbers (p, q) satisfying the condition $\frac{51}{100} < \frac{1}{p} + \frac{1}{q} < \frac{5}{6}$ will be
 (1) 49 (2) 24 (3) 50 (4) 48
10. The number of positive integral values of n for which $(n^3 - 8n^2 + 20n - 13)$ is a prime number is
 (1) 2 (2) 1 (3) 3 (4) 4
11. The number of positive integers ' n ' for which $3n - 4, 4n - 5$ and $5n - 3$ are all primes is
 (1) 1 (2) 2 (3) 3 (4) infinite
12. m, n are natural numbers. The number of pairs (m, n) for which $m^2 + n^2 + 2mn - 2013m - 2013n - 2014 = 0$ is.
 (1) 2000 (2) 2013 (3) 1023 (4) 2007
13. Suppose that a, b, c are distinct numbers such that

$$(b - a)^2 - 4(b - c)(c - a) = 0,$$
 find the value of $\frac{b - c}{c - a}$.
14. Suppose that p and q are prime numbers ($p > q$) and they are roots of the equation $x^2 - 99x + m = 0$ for some m . What is the value of $p - q$?
15. What is the largest possible prime value of $n^2 - 12n + 27$, where n ranges over all positive integers?
16. What is the largest integer that is a divisor of $(n + 1)(n + 3)(n + 5)(n + 7)(n + 9)$ for all positive even integers n ?

17. Let A be the set of positive integers that have no prime factors other than 2, 3 or 5. The infinite sum $\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{8} + \frac{1}{9} + \frac{1}{10} + \frac{1}{12} + \frac{1}{15} + \frac{1}{16} + \frac{1}{18} + \frac{1}{20} + \dots$ of the reciprocals of the elements of A can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. What is $m + n$?
18. For some positive integer n, the number $110n^3$ has 110 positive integer divisors, including 1 and the number $110n^3$. The number $81n^4$ have D positive integer divisors. What is $\frac{D}{5}$?
19. What is the minimum number of digits to the right of the decimal point needed to express the fraction $\frac{123456789}{2^{26} \cdot 5^4}$ as a decimal?
20. How many positive two-digit integers are factors of $2^{24} - 1$?
21. Let N be the positive integer for which the sum of its two smallest factors is 4 and the sum of its two largest factors is 204. Find the value of $\left(\frac{N}{3}\right)$.
22. Let $S = \{1, 2, 3, \dots, 20\}$ be the set of all positive integers from 1 to 20. Suppose that N is the smallest positive integer such that exactly eighteen numbers from S are factors of N, and the only two numbers from S that are not factors of N are consecutive integers. Find the sum of the digits of N.
23. m and n are two positive integers of reverse order (for example 123 and 321) such that $mn = 1446921630$. Now the value of $m + n$ is given as $ab4ba$. Find the product of a and b.
24. Find the number of positive divisors of $(2008^3 + (3 \times 2008 \times 2009) + 1)^2$.
25. Determine the largest prime number less than 5000 of the form $a^n - 1$, where a and n are positive integers, and n is greater than 1.
26. Let f be a real-valued function with the rule $f(x) = x^3 + 3x^2 + 6x + 14$ defined for all real value of x. It is given that a and b are two real numbers such that $f(a) = 1$ and $f(b) = 19$. Find the value of $(a + b)^2$.
27. Positive integers a, b, and 2009, with $a < b < 2009$, form a geometric sequence with an integer ratio. What is a?
28. For $k > 0$, let $I_k = 10 \dots 064$, where there are k zeros between the 1 and the 6. Let $N(k)$ be the number of factors of 2 in the prime factorization of I_k . What is the maximum value of $N(k)$?
29. Suppose that m and n are positive integers such that $75m = n^3$. What is the minimum possible value of $m + n$?
30. Let n be the largest integer that is the product of exactly 3 distinct prime numbers, d, e, and $10d + e$, where d and e are single digits. What is the sum of the digits of n?
31. The digits 1, 2, 3, 4, 5, 6, 7, and 9 are used to form four two-digit prime numbers, with each digit used exactly once. Then the sum of these four prime numbers is given by $10A$. Find the value of A.
32. The positive integers A, B, $A - B$, and $A + B$ are all prime numbers. The sum of these four prime number is
33. Find the number of counterexamples to the statement:
"If N is an odd positive integer the sum of whose digits is 4 and none of whose digits is 0, then N is prime."
34. Three positive integers are each greater than 1, have a product of 27000, and are pairwise relatively prime. What is their mean to the nearest whole number?

35. A positive integer n is nice if there is a positive integer m with exactly four positive divisors (including 1 and m) such that the sum of the four divisors is equal to n . How many numbers in the set $\{2010, 2011, 2012, \dots, 2019\}$ are nice?
36. Let a and b be relatively prime integers with $a > b > 0$ and $\frac{a^3 - b^3}{(a - b)^3} = \frac{73}{3}$. What is $a - b$?

EXERCISE - 2.2

- Given $1x6y7$ is a five-digit number divisible by 9. The number of ordered pairs (x, y) satisfying this is
(1) 5 (2) 6 (3) 8 (4) 11
- The number of three digit numbers that are divisible by 2 but not divisible by 4 is
(1) 200 (2) 225 (3) 250 (4) 450
- a, b, c are the digits of a nine digit number $abcabcabc$. Calculate the quotient when this number is divided by 1001001.
(1) $100a + 10c + b$ (2) $100c + 10b + a$ (3) $100a + 10b + c$ (4) $10a + c$
- Find the number of integers n for which $\frac{n}{20-n}$ is the square of an integer.
(1) 1 (2) 2 (3) 3 (4) 4
- The years of 20th century and 21th century are of 4 digits. The number of years which are divisible by the product of the four digits of the year is
(1) 7 (2) 8 (3) 9 (4) none of these
- The number of numbers of the form $30a0b03$ that are divisible by 13, where a, b are digits, is
(1) 5 (2) 6 (3) 7 (4) 0
- A natural number n has exactly two divisors and $(n + 1)$ has three divisors. The number of divisors of $(n + 2)$ is
(1) 2 (2) 3 (3) 4 (4) depends on the value of n
- If $n = 10^{10} - 1$, the number of digits in n^3 is
(1) 30 (2) 28 (3) 32 (4) 27
- The last two digits of 3^{2012} , when represented in decimal notation, will be
(1) 81 (2) 01 (3) 41 (4) 21
- Give the units digit of the number 3^{2011} .
(1) 7 (2) 8 (3) 10 (4) 11
- When written out completely 16^{2011} has m digits and 625^{2011} has n digits. Find the value of $(m + n)$.
(1) 8000 (2) 4045 (3) 8045 (4) 7040
- Find the units digit of $(2013)^{2013}$.
(1) 2 (2) 3 (3) 5 (4) 7
- Suppose that n is the product of three consecutive integers and that n is divisible by 7. Which of the following is not necessarily a divisor of n ?
(1) 14 (2) 21 (3) 28 (4) 6
- Find the last two digits of $9^{100} - 1$.
- Find the sum of digits of the largest positive integer n such that $n!$ ends with exactly 100 zeros.

16. Find the number of positive integers n such that

$$n + 2n^2 + 3n^3 + \dots + 2005n^{2005}$$

is divisible by $n - 1$.

17. Let n be the number

$$\underbrace{(999\ 999\ 999\dots 999)^2}_{2006\ 9\text{'s}} - \underbrace{(666\ 666\ 666\dots 666)^2}_{2006\ 6\text{'s}}.$$

Find the remainder when n is divided by 11.

18. What is the sum of all the digits of the largest positive integer n for which $n^3 + 2006$ is divisible by $n + 26$?
19. Find the smallest positive integer n such that $n(n + 1)(n + 2)$ is divisible by 247.
20. Total number of 0's at the end of the value of the product $1 \times 2 \times 3 \times 4 \times \dots \times 2008$ is N , then find the value of $\sqrt{n + 125}$
21. Let n denote the smallest positive integer that is divisible by both 4 and 9, and whose base-10 representation consists of only 4's and 9's, with at least one of each. What will be the sum of the last four digits of n ?
22. An n -digit positive integer is cute if its n digits are an arrangement of the set $\{1, 2, \dots, n\}$ and its first k digits form an integer that is divisible by k , for $k = 1, 2, \dots, n$. For example, 321 is a cute 3-digit integer because 1 divides 3, 2 divides 32, and 3 divides 321. How many cute 6-digit integers are there?
23. Consider the non-decreasing sequence of positive integers
 $1, 2, 2, 3, 3, 3, 4, 4, 4, 4, 5, 5, 5, 5, \dots$
 in which the n^{th} positive integer appears n times. The remainder when the 2019th term is divided by 5 is
24. Mary chose an even 4-digit number n . She wrote down all the divisors of n in increasing order from left to right: $1, 2, \dots, \frac{n}{2}, n$. At some moment Mary wrote 323 as a divisor of n . If the smallest possible value of the next divisor written to the right of 323 is N . Find the value of $\frac{N}{10}$.
25. The largest divisor of 2, 014, 000, 000 is itself. If its fifth largest divisor is X . Find sum of all the digits of X .
26. How many three-digit numbers are not divisible by 5, have digits that sum to less than 20, and have the first digit equal to the third digit?
27. How many even integers are there between 200 and 700 whose digits are all different and come from the set $1, 2, 5, 7, 8, 9$?

EXERCISE - 2.3

1. The sum of the digits of the number $10^n - 1$ is 3798. The value of n is
 (1) 431 (2) 673 (3) 422 (4) 501
2. a, b, c, d are natural numbers such that $a = bc, b = cd, c = da$ and $d = ab$. Then $(a + b)(b + c)(c + d)(d + a)$ is equal to
 (1) $(a + b + c + d)^2$ (2) $(a + b)^2 + (c + d)^2$ (3) $(a + d)^2 + (b + c)^2$ (4) $(a + c)^2 + (b + d)^2$
3. A three digit number with digits A, B, C in that order is divisible by 9. A is an odd digit and C is an even digit. B and C are non zero. The number of such three digit numbers is
 (1) 4 (2) 8 (3) 16 (4) 20

4. Nine numbers are written in ascending order. The middle number is also the average of the nine numbers. The average of the 5 larger numbers is 68 and the average of the 5 smaller numbers is 44. The sum of all the numbers is
 (1) 540 (2) 450 (3) 504 (4) 501
5. If the average of 20 different positive integers is 20 then the greatest possible number among these 20 numbers can be
 (1) 210 (2) 200 (3) 190 (4) 180
6. If a, b are natural numbers such that $a + b = 2008$, then $(-1)^a + (-1)^b$ is
 (1) 1 (2) -1 (3) 2 (4) 2 or -2
7. The number 16^{12} is obtained from the number 8^4 by raising the smaller number to the power n . Then n is
 (1) 3 (2) 4 (3) $\frac{16}{3}$ (4) $\frac{8}{3}$
8. The remainder when the number $(2 \times 3 \times 4 \times 2012 \times 2013 \times 2014 \times 2015) - 2008$ is divided by 2015 is
 (1) 1 (2) 2008 (3) 7 (4) 2007
9. The last digit in the finite decimal representation of the number $\left(\frac{1}{5}\right)^{2004}$ is
 (1) 8 (2) 6 (3) 4 (4) 2
10. A natural number n has exactly two divisors and $(n+1)$ has three divisors. The number of divisors of $(n+2)$ is
 (1) 2 (2) 3 (3) 4 (4) depends on the value of n
11. The sum of two natural number is 100. Which of the following can be the greatest product of the numbers.
 (1) 1000 (2) 4000 (3) 2500 (4) 900
12. A computer is printing a list of the seventh powers of all natural numbers, that is the sequence $1^7, 2^7, 3^7, \dots$. The number of terms (or numbers) between 5^{21} and 4^{28} are
 (1) 12 (2) 130 (3) 14 (4) 150
13. For a positive integer k , we write

$$(1+x)(1+2x)(1+3x)\cdots(1+kx) = a_0 + a_1x + a_2x^2 + \cdots + a_kx^k,$$
 where $a_0, a_1, \dots, a_k$ are the coefficients of the polynomial. Find the sum of all the digits of smallest possible value of k if $a_0 + a_1 + a_2 + \cdots + a_{k-1}$ is divisible by 2005.
14. The smallest three-digit number n such that if the three digits are a, b and c , then

$$n = a + b + c + ab + bc + ac + abc.$$
 Evaluate $\frac{n+1}{5}$
15. What is the largest positive integer n satisfying $n^{200} < 5^{300}$?
16. N pieces of candy are made and packed into boxes, with each box containing 45 pieces. If N is a non-zero perfect cube and 45 is one of its factors, what is the least possible number of boxes that can be packed?
17. For any non-empty finite set A of real numbers, let $s(A)$ be the sum of the elements in A . There are exactly 61 3-element subsets A of $\{1, \dots, 23\}$ with $s(A) = 36$. The total number of 3-element subsets of $\{1, \dots, 23\}$ with $s(A) < 36$ is given by N . Then evaluate $\frac{N+45}{25}$

18. Let the sum of the digits of the given product is written as A8B6C. Find $A + B + C$

$$\underbrace{(111111111 \dots 111)}_{2007 \text{ 1's}} \times 2007$$

19. When 2007 bars of soap are packed into N boxes of equal size, where N is an integer strictly between 200 and 300, there are extra 5 bars remaining. Find last two digit of N .
20. Let the number of distinct integers among the numbers is N . Find the sum of all digits of N .

$$\left\lceil \frac{1^2}{2007} \right\rceil, \left\lceil \frac{2^2}{2007} \right\rceil, \dots, \left\lceil \frac{2007^2}{2007} \right\rceil.$$

21. For each integer $n \geq 4$, let a_n denote the base- n number $0.\overline{133}_n$. The product $a_4 a_5 \dots a_{99}$ can be expressed as $\frac{m}{n!}$, where m and n are positive integers and n is as small as possible. What is the value of sum of the digits of m ?

22. The square

is a multiplicative magic square. That is, the product of the numbers in each row, column, and diagonal is the same. If all the entries are positive integers, what is the sum of the possible values of g ?

50	b	c
d	e	f
g	h	2

23. How many four-digit numbers N have the property that the three-digit number obtained by removing the leftmost digit is one ninth of N ?

24. If a , b , and c are positive real numbers such that $a(b+c) = 152$, $b(c+a) = 162$, and $c(a+b) = 170$, then $\frac{abc}{12}$ is

25. If x , y , and z are positive numbers satisfying
 $x + 1/y = 4$, $y + 1/z = 1$, and $z + 1/x = 7/3$,
 then find xyz ?

26. There are x integer values of n (not necessarily positive) so that value of $4000 \cdot \left(\frac{2}{5}\right)^n$ is an integer. What is x ?

27. Let S be a set of 6 integers taken from $\{1, 2, \dots, 12\}$ with the property that if a and b are elements of S with $a < b$, then b is not a multiple of a . What is the square of the least possible value of an element in S ?

28. R nonnegative integers can be written in the form

$$a_7 \cdot 3^7 + a_6 \cdot 3^6 + a_5 \cdot 3^5 + a_4 \cdot 3^4 + a_3 \cdot 3^3 + a_2 \cdot 3^2 + a_1 \cdot 3^1 + a_0 \cdot 3^0,$$

where $a_i \in \{-1, 0, 1\}$ for $0 \leq i \leq 7$. What is the last two digits of R ?

29. How many odd positive 3-digit integers are divisible by 3 but do not contain the digit 3?

30. Let p and q be positive integers such that $\frac{5}{9} < \frac{p}{q} < \frac{4}{7}$ and q is as small as possible. What is $(q-p)^2$?

31. A function f is defined recursively by $f(1) = f(2) = 1$ and $f(n) = f(n-1) - f(n-2) + n$ for all integers $n \geq 3$. What is the sum of the digits of $f(2018)$?

32. Sunita has 30 thin rods, one each of every integer length from 1 cm through 30 cm. She places the rods with lengths 3 cm, 7 cm, and 15 cm on a table. She then wants to choose a fourth rod that she can put with these three to form a quadrilateral with positive area. How many of the remaining rods can she choose as the fourth rod?
33. A five-digit palindrome is a positive integer with respective digits $abcba$, where a is non-zero. Let S be the sum of all five-digit palindromes. What is the sum of the digits of S ?
34. The products $(8)(888\dots 8)$, where the second factor has k digits, is an integer whose digits have a sum of 1000. What is the sum of digits of k ?
35. There are total P positive integers value of n , so that $\frac{n}{30-n}$ is also a positive integer. What is P^2 ?
36. Let S be the set of positive integers n for which $\frac{1}{n}$ has the repeating decimal representation $0.\overline{ab} = 0.ababab\dots$, with a and b different digits. The sum of the elements of S is a three digit number abc . What is the value of $\frac{abc}{11}$?
37. Let a , b , and c be digit with $a \neq 0$. The three-digit integer abc lies one third of the way from the square of a positive integer to the square of the next larger integer. The integer acb lies two thirds of the way between the same two squares. What is $a + b + c$?
38. There are p pairs of positive integers (a, b) such that $\gcd(a, b) = 1$ and $\frac{a}{b} + \frac{14b}{9a}$ is an integer. What is $7 \times p$?
39. Let a, b, c, d, e, f, g and h be distinct elements in the set $\{-7, -5, -3, -2, 2, 4, 6, 13\}$. What is the minimum possible value of $(a + b + c + d)^2 + (e + f + g + h)^2$?
40. There are 88 numbers $a_1, a_2, a_3, \dots, a_{88}$ and each of them is either equals to -3 or -1 . Given that $a_1^2 + a_2^2 + \dots + a_{88}^2 = 280$, and $a_1^4 + a_2^4 + \dots + a_{88}^4 = 10^3a + b$. Find the product of a and b .
41. The difference between the highest common factor and the lowest common multiple of x and 18 is 120. Find the value of x .
42. Given that n is a ten-digit number in the form $\overline{2007x2008y}$ where x and y can be any of the digits 0, 1, 2, ..., 9. How many such numbers n are there that are divisible by 33?
43. How many positive integers n , where $10 \leq n \leq 100$, are there such that $\frac{n^2 - 9}{n^2 - 7}$ is a fraction in its lowest terms?
44. Let n be a positive integer such that $n^2 + 19n + 48$ is a perfect square. Find the value of n .
45. Find the smallest integer n such that $n(\sqrt{101} - 10) > 1$.
46. Find the smallest positive integer N such that $2^n > n^2$ for every integer n in $\{N, N+1, N+2, N+3, N+4\}$.
47. The highest natural number which are divisible by 30 and have exactly 30 different positive divisors can be written as $A1B5C$ then find $A + B + C$.
48. Let p and q represent two consecutive prime numbers. For some fixed integer n , the set $\{n-1, 3n-19, 38-5n, 7n-45\}$ represents $\{p, 2p, q, 2q\}$, but not necessarily in that order. Find the value of n .

49. If a , b and c are positive real numbers such that
 $ab + a + b = bc + b + c = ca + c + a = 35$,
 then the value of $(a + 1)(b + 1)(c + 1) = x^y$. Find the product of x and y .
50. Find the largest possible value of n such that there exist n consecutive positive integers whose sum is equal to 2010.
51. Let S be the set of all integers n such that $\frac{8n^3 - 96n^2 + 360n - 400}{2n - 7}$ is an integer. Find the value of $\sum_{n \in S} |n|$.
52. Find the sum of all positive integers p such that the expression $(x - p)(x - 13) + 4$ can be expressed in the form $(x + q)(x + r)$ for distinct integers q and r .
53. Let $P_k = 1 + \frac{1}{k} - \frac{1}{k^2} - \frac{1}{k^3}$, where k is a positive integer. Find the sum of all the digits of least positive integer n such that the product $P_2 P_3 \cdots P_n$ exceeds 2010.
54. What is the sum of the digits of the square of 111, 111, 111?
55. Let a , b , c , and d be real numbers with $|a - b| = 2$, $|b - c| = 3$ and $|c - d| = 4$. What is the sum of all possible values of $|a - d|$?
56. Let $k = 2008^2 + 2^{2008}$. What is the units digit of $k^2 + 2^k$?
57. The numbers from 1 to 8 are placed at the vertices of a cube in such a manner that the sum of the four numbers on each face is the same. What is this common sum?
58. The saxena family consists of a mother, a father, and some children. The average age of the members of the family is 20, the father is 48 years old, and the average age of the mother and children is 16. How many children are in the family?
59. How many ordered pairs (m, n) of positive integers, with $m > n$, have the property that their squares differ by 96?
60. For each positive integer n , let $S(n)$ denote the sum of the digits of n . For how many values of n is $n + S(n) + S(S(n)) = 2007$?
61. If N is the number of four-digit positive integers have at least one digit that is a 2 or a 3 then find the sum of the digits present in N ?
62. How many three-digit numbers satisfy the property that the middle digit is the average of the first and the last digits?
63. How many positive integer cubes divide $3! \cdot 5! \cdot 7!$?
64. The sum of the digits of a two-digit number is subtracted from the number. The units digit of the result is 6. How many two-digit numbers have this property?
65. For how many positive integers n does $1 + 2 + \cdots + n$ evenly divide from $6n$?
66. For each positive integer $m > 1$, let $P(m)$ denote the greatest prime factor of m . For how many positive integers n is it true that both $P(n) = \sqrt{n}$ and $P(n + 48) = \sqrt{n + 48}$?
67. Suppose that $4^a = 5$, $5^b = 6$, $6^c = 7$ and $7^d = 8$. What is $\left(\frac{a \cdot b \cdot c \cdot d}{2}\right)$?
68. All of David's telephone numbers have the form 555 - abc - defg, where a , b , c , d , e , f and g are distinct digits and in increasing order and none is either 0 or 1. How many different telephone numbers can David have?

69. For how many positive integers n less than or equal to 24 is $n!$ evenly divisible by $1 + 2 + \dots + n$?
70. Let x and y be two-digit integers such that y is obtained by reversing the digits of x . Suppose that the integers x and y satisfy $x^2 - y^2 = m^2$ for some positive integer ' m '. What is the value of $\left(\frac{x+y+m}{11}\right)$?
71. A subset ' B ' of the set of integers from 1 to 100 inclusive, has the property that no two elements of B sum to 125. What is the maximum possible number of elements in B ?
72. Let n be a 5-digit number, and let q and r be the quotient and remainder, respectively, when n is divided by 100. Let the total number of value of n for which $q + r$ divisible by 11 is P . Then find the last two digits of P .
73. Let $F(x)$ denote the sum of the digits of the positive integer x . For example, $F(8) = 8$ and $F(123) = 1 + 2 + 3 = 6$. For how many two-digit values of x is $F(F(x)) = 3$?
74. Given that $3^8 \cdot 5^2 = a^b$, where both a and b are positive integers, find the smallest possible value for $\left(\frac{a+b+93}{25}\right)$.
75. What is the largest integer that is a divisor of $(n+1)(n+3)(n+5)(n+7)(n+9)$ for all positive even integers n ?
76. The product of three consecutive positive integers is 8 times their sum. What is the sum of their squares?
77. Let $P(n)$ and $S(n)$ denote the product and the sum, respectively, of the digits of the integer n . For example, $P(23) = 6$ and $S(23) = 5$. Suppose N is a two-digit number such that $N = P(N) + S(N)$. What is the units digit of N ?
78. The mean of three numbers is 10 more than the least of the numbers and 15 less than the greatest. The median of the three numbers is 5. What is their sum?
79. Two non-zero real numbers, a and b , satisfy $ab = a - b$. Then find the possible value of $\frac{a}{b} + \frac{b}{a} - ab$?
80. There are N number of positive integers not exceeding 2001 & are multiples of 3 or 4 but not 5. Then find the value of $\left(\frac{N-1}{50}\right)$.
81. In the magic square shown, the sums of the numbers in each row, column, and diagonal are the same. Five of these numbers are represented by v, w, x, y , and z . Find $y + z$.

v	24	w
18	x	y
25	z	21

82. A subset of the integers $1, 2, \dots, 100$ has the property that none of its members is 3 times another. What is the largest number of members such a subset can have?
83. A positive integer N is a palindrome if the integer obtained by reversing the sequence of digits of N is equal to N . The year 1991 is the only year in the current century with the following two properties:
- (a) It is a palindrome.
- (b) It factors as a product of a 2-digit prime palindrome and a 3-digit prime palindrome.
- How many years in the millenium between 1000 and 2000 (including the year 1991) have properties (a) and (b)?

84. The two-digit integers from 19 to 92 are written consecutively to form the large integer

$$N = 19202122\dots 909192$$

If 3^k is the highest power of 3 that is a factor of N , then $k =$

85. For how many n in $\{1, 2, 3, \dots, 100\}$ is the tens digit of n^2 odd?

86. If a and b are digits for which

$$\begin{array}{r} 2a \\ \times b3 \\ \hline 69 \\ 92 \\ \hline 989 \end{array}$$

then $a + b =$

87. For positive integers n , denote by $D(n)$ the number of pairs of different adjacent digits in the binary (base two) representation of n . For example, $D(3) = D(11_2) = 0$, $D(21) = D(10101_2) = 4$, and $D(97) = D(1100001_2) = 2$. For how many positive integers n less than or equal to 97 does $D(n) = 2$?

88. Call a 7-digit telephone number $d_1d_2d_3d_4d_5d_6d_7$ memorable if the prefix sequence $d_1d_2d_3$ is exactly the same as either of the sequences $d_4d_5d_6$ or $d_5d_6d_7$ (possibly both). Assuming that each d_i can be any of the ten decimal digits 0, 1, 2, ..., 9. Suppose the total no. of different memorable telephone no.s is ' n '. Then find the sum of digits of ' n '.

89. The number of ordered pairs of integers (m, n) for which $mn \geq 0$ and

$$m^3 + n^3 + 99mn = 33^3$$

is equal to

90. All even numbers from 2 to 98 inclusive, except those ending in 0, are multiplied together. What is the rightmost digit (the units digit) of the product?
91. Given that $xyz = 2014$, and x, y and z are positive integers such that $x < y < z$, how many possible triples (x, y, z) are there?
92. In the following equation, each of the letters represents uniquely a different digit in base ten :

$$(YE).(ME) = TTT$$

The sum $E + M + T + Y$ equals

93. Let a, b, c , and d be positive integers such that $\gcd(a, b) = 24$, $\gcd(b, c) = 36$, $\gcd(c, d) = 54$, and $70 < \gcd(d, a) < 100$. Which of the following must be a divisor of a ?
94. For a positive integer n and nonzero digits a, b , and c , let A_n be the n -digit integer each of whose digits is equal to a ; let B_n be the n -digit integer each of whose digits is equal to b , and let C_n be the $2n$ -digit (not n -digit) integer each of whose digits is equal to c . What is the greatest possible value of $a + b + c$ for which there are at least two values of n such that $C_n - B_n = A_n^2$?
95. How many ordered pairs (a, b) of positive integers satisfy the equation

$$a \cdot b + 63 = 20 \cdot \text{lcm}(a, b) + 12 \cdot \gcd(a, b),$$

where $\gcd(a, b)$ denotes the greatest common divisor of a and b , and $\text{lcm}(a, b)$ denotes their least common multiple?

96. There are 10 horses, named Horse 1, Horse 2, ..., Horse 10. They get their names from how many minutes it takes them to run one lap around a circular race track: Horse k runs one lap in exactly k minutes. At time 0 all the horses are together at the starting point on the track. The horses start running in the same direction, and they keep running around the circular track at their constant speeds. The least time $S > 0$, in minutes, at which all 10 horses will again simultaneously be at the starting point is $S = 2520$. Let $T > 0$ be the least time, in minutes, such that at least 5 of the horses are again at the starting point. What is the sum of the digits of T ?

97. How many ordered triples (x, y, z) of positive integers satisfy $\text{lcm}(x, y) = 72$, $\text{lcm}(x, z) = 600$ and $\text{lcm}(y, z) = 900$?
98. For the greatest value of x , if 2^x is a factor of $10^{1002} - 4^{501}$. Find the value of $\frac{x}{15}$.
99. In base 10, the number 2013 ends in the digit 3. In base 9, on the other hand, the same number is written as $(2676)_9$ and ends in the digit 6. For how many positive integers b does the base- b representation of 2013 end in the digit 3?
100. Real numbers x and y satisfy the equation $x^2 + y^2 = 10x - 6y - 34$. What is $x + y$?
101. The number 2013 has the property that its units digit is the sum of its other digits, that is $2 + 0 + 1 = 3$. How many integers less than 2013 but greater than 1000 share this property?
102. The number 2013 is expressed in the form

$$2013 = \frac{a_1!a_2! \dots a_m!}{b_1!b_2! \dots b_n!},$$

where $a_1 \geq a_2 \geq \dots \geq a_m$ and $b_1 \geq b_2 \geq \dots \geq b_n$ are positive integers and $a_1 + b_1$ is as small as possible. What is $|a_1 - b_1|$?

103. Two integers have a sum of 26. When two more integers are added to the first two integers the sum is 41. Finally when two more integers are added to the sum of the previous four integers the sum is 57. What is the minimum number of even integers among the 6 integers?
104. How many ordered pairs of positive integers (M, N) satisfy the equation $\frac{M}{6} = \frac{6}{N}$?
105. Geeta and Babeeta play the following game. An integer between 0 and 999, inclusive, is selected and given to Geeta. Whenever Geeta receives a number, he doubles it and passes the result to Babeeta. Whenever Babeeta receives a number, she adds 50 to it and passes the result to Geeta. The winner is the last person who produces a number less than 1000. Let N be the smallest initial number that results in a win for Geeta. What is the sum of the digits of N ?
106. A majority of the 30 students in Mrs. Bhatnagar's class bought pencils at the school bookstore. Each of these students bought the same number of pencils, and this number was greater than 1. The cost of a pencil in paisa was greater than the number of pencils each student bought, and the total cost of all the pencils was ₹ 17.71. What was the cost of a pencil in paisa?

EXERCISE - 2.4

1. Simplify $\frac{1 + \sqrt{2} + \sqrt{3}}{1 - \sqrt{2} + \sqrt{3}}$ by rationalizing the denominator.
2. Simplify the expression $\frac{x^2 - 4x + 3 + (x - 1)\sqrt{x^2 - 9}}{x^2 + 4x + 3 + (x - 1)\sqrt{x^2 - 9}}$, where $x > 3$.
3. Evaluate $(\sqrt{5} + \sqrt{6} + \sqrt{7})(\sqrt{5} + \sqrt{6} - \sqrt{7})(\sqrt{5} - \sqrt{6} + \sqrt{7})(-\sqrt{5} + \sqrt{6} + \sqrt{7})$.
4. Suppose that $a = \sqrt{6} - 2$ and $b = 2\sqrt{2} - \sqrt{6}$. Then
(A) $a > b$, (B) $a = b$, (C) $a < b$, (D) $b = \sqrt{2a}$, (E) $a = \sqrt{2b}$.

5. Arrange the three values $a = \sqrt{27} - \sqrt{26}$, $b = \sqrt{28} - \sqrt{27}$, $c = \sqrt{29} - \sqrt{28}$ in ascending order.
6. The number of integers x which satisfies the inequality $\frac{3}{1+\sqrt{3}} < x < \frac{3}{\sqrt{5}-\sqrt{3}}$ is
(A) 2, (B) 3, (C) 4, (D) 5, (E) 6.
7. Calculate the value of $\frac{1}{1-\sqrt[4]{5}} + \frac{1}{1+\sqrt[4]{5}} + \frac{2}{1+\sqrt{5}}$.
8. Given $a > b > c > d > 0$, and $U = \sqrt{ab} + \sqrt{cd}$, $V = \sqrt{ac} + \sqrt{bd}$, $W = \sqrt{ad} + \sqrt{bc}$. Use "<" to connect U, V, W.
9. Simplify $\sqrt[3]{3} \left(\sqrt[3]{\frac{4}{9}} - \sqrt[3]{\frac{2}{9}} + \sqrt[3]{\frac{1}{9}} \right)^{-1}$.
10. Given $a = \sqrt[3]{4} + \sqrt[3]{2} + 1$, find the value of $\frac{3}{a} + \frac{3}{a^2} + \frac{1}{a^3}$.
11. Given $x + y = \sqrt{3\sqrt{5} - \sqrt{2}}$, $x - y = \sqrt{3\sqrt{2} - \sqrt{5}}$, find the value of xy .
12. Evaluate $\sqrt{a+3+4\sqrt{a-1}} + \sqrt{a+3-4\sqrt{a-1}}$.
13. Simplify $\sqrt[3]{\frac{(\sqrt{a-1}-\sqrt{a})^5}{\sqrt{a-1}+\sqrt{a}}} + \sqrt[3]{\frac{(\sqrt{a-1}+\sqrt{a})^5}{\sqrt{a}-\sqrt{a-1}}}$.
14. Simplify $\sqrt{1+a^2} + \sqrt{1+a^2+a^4}$.
15. Given $\sqrt{x} = \sqrt{a} - \frac{1}{\sqrt{a}}$, find the value of $\frac{x+2+\sqrt{x^2+4x}}{x+2-\sqrt{x^2+4x}}$.
16. Find the last two digits of the greatest integer less than $(2 + \sqrt{3})^4$.
17. Consider the function $f(x) = \frac{1}{3^x + \sqrt{3}}$. Find the value of $\sqrt{3} [f(-5) + f(-4) + f(-3) + f(-2) + f(-1) + f(0) + f(1) + f(2) + f(3) + f(4) + f(5) + f(6)]$.
18. How many ordered pairs of positive integers (x, y) satisfy the equation $x\sqrt{y} + y\sqrt{x} + \sqrt{2006xy} - \sqrt{2006x} - \sqrt{2006y} - 2006 = 0$?
19. Evaluate X in its simplest form then find sum of all digits of X . Where X is given as $X = \sqrt{2008 + 2007\sqrt{2008 + 2007\sqrt{2008 + 2007\sqrt{\dots}}}}$.
20. Find the number of ordered pairs of positive integers (x, y) that satisfy the equation $x\sqrt{y} + y\sqrt{x} + \sqrt{2009xy} - \sqrt{2009x} - \sqrt{2009y} - 2009 = 0$.
21. Evaluate $\sqrt[3]{77-20\sqrt{13}} + \sqrt[3]{77+20\sqrt{13}}$.

22. Find the value of $(25 + 10\sqrt{5})^{1/3} + (25 - 10\sqrt{5})^{1/3}$.

23. Let $a = \frac{1 + \sqrt{2009}}{2}$. Find the value of $(a^3 - 503a - 500)^5$.

24. Let n be the positive integer such that

$$\frac{1}{9\sqrt{11} + 11\sqrt{9}} + \frac{1}{11\sqrt{13} + 13\sqrt{11}} + \frac{1}{13\sqrt{15} + 15\sqrt{13}} + \dots + \frac{1}{n\sqrt{n+2} + (n+2)\sqrt{n}} = \frac{1}{9}$$

Find the value of n .

25. If $x = \frac{1}{2} \left(\sqrt[3]{2009} - \frac{1}{\sqrt[3]{2009}} \right)$ and $y = (x + \sqrt{1+x^2})^3$, then find the sum of the digits of y .

26. Let x and y be real numbers satisfying

$$y = \sqrt{\frac{2008x+2009}{2010x-2011}} + \sqrt{\frac{2008x+2009}{2011-2010x}} + 25.$$

Find the value of y .

27. If the value of fraction $\frac{2(\sqrt{2}) + \sqrt{6}}{3\sqrt{2} + \sqrt{3}}$ is equal to $12P$. Find P .

28. Let $T = \frac{1}{3-\sqrt{8}} - \frac{1}{\sqrt{8}-\sqrt{7}} + \frac{1}{\sqrt{7}-\sqrt{6}} - \frac{1}{\sqrt{6}-\sqrt{5}} + \frac{1}{\sqrt{5}-2}$; then value of T is

29. Suppose that real number x satisfies $\sqrt{49-x^2} - \sqrt{25-x^2} = 3$.

What is the value of $\sqrt{49-x^2} + \sqrt{25-x^2}$?

30. If the value of $\sqrt{9-6\sqrt{2}} + \sqrt{9+6\sqrt{2}}$ is k , find $\frac{25}{\sqrt{6}} \times k$.

EXERCISE - 2.5

1. Prove that $7 \mid (2222^{5555} + 5555^{2222})$.
2. Find the remainder of $47^{37^{27}}$ when it is divided by 11.
3. What is the remainder when 9^{1990} is divided by 11?
4. When a positive integer n is divided by 5, 7, 9, 11, the remainders are 1, 2, 3, 4 respectively. Find the minimum value of n .
5. Find the last two digits of 2^{999} .
6. Find the remainder of $(257^{33} + 46)^{26}$ when it is divided by 50.
7. What is the smallest positive integer $n > 1$ such that 3^n ends with 003?
8. Find all the positive integers n , such that $n \cdot 2^n + 1$ is divisible by 3.
9. Prove that when $\overline{abc}$ is a multiple of 37, then so is the number $\overline{bca}$.
10. Find all positive integers with initial digit 6 such that the integer formed by deleting this 6 is $1/25$ of the original integer.

11. Let x be a 3-digit number such that the sum of the digits equals 21. If the digits of x are reversed, the number thus formed exceeds x by 495. What is x ?
12. Given that a four digit number n and the sum of all digits of n have a sum 2001. Find n .
13. When $N = \underbrace{11\dots11}_{1989 \text{ digits}} \times \underbrace{11\dots11}_{1989 \text{ digits}}$, what is the sum of all digits of N ?
14. Find the maximum value of the ratio of three digit number to the sum of its digits.
15. Find all the three digit numbers $n = \overline{abc}$ such that $n = (a + b + c)^3$.
16. Given that the natural numbers a, b, c are formed by the same n digits x, n digits y , and $2n$ digits z respectively. For any $n \geq 2$ find the digits x, y, z such that $a^2 + b = c$.
17. When a two digit number is divided by the number formed by exchanging the two digits, the quotient is equal to its remainder. Find the two digit number.
18. Find the maximum perfect square, such that after deleting its last two digits (which is assumed to be not all zeros), the remaining part is still a perfect square.
19. Find the last two digits (in order) of $7^{3^{2005}}$.
20. Find the smallest two-digit number N such that the sum of digits of $10^N - N$ is divisible by 170.
21. Let $N = 12345678910111213\dots4344$ be the 79 digit number obtained that is formed by writing the integers from 1 to 44 in order, one after the other. The remainder when N is divided by 45 is r . What is r^2 ?
22. Let $k = 2008^2 + 2^{2008}$. What is the square of the units digit of $k^2 + 2^k$?
23. Anna randomly picked five integers from the following list
53, 62, 66, 68, 71, 82, 89
and discover that the average value of the five integers she picked is still an integer. If two of the integers she picked were 62 and 89, find the mean of the remaining three integers.
24. Find the remainder when $(1! \times 1) + (2! \times 2) + (3! \times 3) + \dots + (286! \times 286)$ is divided by $2^4 \times 5^3$.
25. What is the remainder when $3^0 + 3^1 + 3^2 + \dots + 3^{2009}$ is divided by 8?
26. How many distinct four-digit numbers are divisible by 3 and have 23 as their last two digits?
27. What is the last two digits of $2015^{2016} - 2017$?

EXERCISE - 2.6

1. If $(x - 1)(x + 3)(x - 4)(x - 8) + m$ is a perfect square, then m is
(1) 32, (2) 24, (3) 98, (4) 196.
2. If $n + 20$ and $n - 21$ are both perfect squares, where n is a natural number, find n .
3. Find the maximal integer x such that $4^{27} + 4^{10000} + 4^x$ is a perfect square.
4. Prove that for any positive integer n , $n^4 + 2n^3 + 2n^2 + 2n + 1$ is not a perfect square.
5. Prove that there is no three digit number $\overline{abc}$, such that $\overline{abc} + \overline{bca} + \overline{cab}$ is a perfect square.
6. Prove that the equation $a^2 + b^2 - 8c = 6$ has no integer solution.

7. Given that p is a prime number, and the sum of all positive divisors of p^4 is a perfect square. Find the number of such primes p .
8. If x and y are positive integers, prove that the values of $x^2 + y + 1$ and $y^2 + 4x + 3$ cannot both be perfect squares at the same time.
9. Let d be any positive integer not equal to 2, 5, or 13. Show that one can find distinct a, b in the set $\{2, 5, 13, d\}$ such that $ab - 1$ is not a perfect square.
10. Prove that the number $3^n + 2 \times 17^n$, where n is a non-negative integer, is never a perfect square.
11. Let p be a prime number such that the next larger number is a perfect square. Find the sum of all such prime numbers. (For example, if you think that 11 and 13 are two such prime numbers, then the sum is 24.)
12. The value of greatest integer N such that

$$N \leq \sqrt{2007^2 - 20070 + 31}.$$

Find $\frac{N-2}{100}$

13. Find the sum of all the digit of largest integer N such that both $N + 496$ and $N + 224$ are perfect squares.
14. The sum of 18 consecutive positive integers is a perfect square. In the smallest possible value of this sum, find the sum of first two integers?
15. Let n be the smallest positive integer such that n is divisible by 20, n^2 is a perfect cube, and n^3 is a perfect square. The number n contains N digits. What is N^2 ?
16. How many positive integers n are there such that $7n + 1$ is a perfect square and $3n + 1 < 2008$?
17. The product $1 \times 2 \times 3 \times \dots \times n$ is denoted by $n!$. For example $4! = 1 \times 2 \times 3 \times 4 = 24$. Let $M = 1! \times 2! \times 3! \times 4! \times 5! \times 6! \times 7! \times 8! \times 9!$. Now the number of factors of M are perfect squares is (21b). Then find the value of b .
18. Find the number of ordered pairs (x, y) , where x is an integer and y is a perfect square, such that $y = (x - 90)^2 - 4907$.
19. The number $25^{64} \cdot 64^{25}$ is the square of a positive integer N . In decimal representation, the sum of the digits of N is.
20. If 554 is the base b representation of the square of the number whose base b representation is 24, then b , when written in base 10, equals

EXERCISE - 2.7

1. If n is a positive integer, prove that $\left[\sqrt{n} + \sqrt{n+1} \right] = \left[\sqrt{4n+2} \right]$.
2. Solve equation $[x^3] + [x^2] + [x] = [x] - 1$.
3. Solve equation $[3x + 1] = 2x - \frac{1}{2}$, and find the sum of all roots.
4. Find the minimum natural number n , such that the equation $\left[\frac{10^n}{x} \right] = 1989$ has integer solution x .
5. How many of the first 1000 positive integers can be expressed in the form $[2x] + [4x] + [6x] + [8x]$, where x is a real number, and $[z]$ denotes the greatest integer less than or equal to z ?
6. How many different non-negative integers are there in the sequence $\left[\frac{1^2}{1980} \right], \left[\frac{2^2}{1980} \right], \left[\frac{3^2}{1980} \right], \dots, \left[\frac{1980^2}{1980} \right]$?

7. For every positive integer n , show that $\lceil \sqrt{n} + \sqrt{n+1} \rceil = \lceil \sqrt{4n+1} \rceil = \lceil \sqrt{4n+2} \rceil = \lceil \sqrt{4n+3} \rceil$.
8. Solve the equation $\{x\} + x = 2\{x\} + 10$.
9. Find the last two digits of the number $\left\lceil \frac{10^{93}}{10^{31} + 3} \right\rceil$ (Write down the tens digit first, then write down the units digit).
10. Solve the equation $x + \frac{92}{x} = \{x\} + \frac{92}{\{x\}}$.
11. If x is a real number that satisfies

$$\left\lceil x + \frac{11}{100} \right\rceil + \left\lceil x + \frac{12}{100} \right\rceil + \dots + \left\lceil x + \frac{99}{100} \right\rceil = 765,$$

find the value of $\{900\} - \{100x\}$. Here $\{a\}$ denotes the largest integer $\leq a$.

12. Find the number of consecutive 0's at the end of the base 10 representation of $206!$.
13. There are P number of positive integer n less than 2007 such that $\left\lceil \frac{n}{2} \right\rceil + \left\lceil \frac{n}{3} \right\rceil + \left\lceil \frac{n}{6} \right\rceil = n$ where $\{x\}$ is the greatest integer less than or equal to x . Write the last two digit of P .
(For example, $\{2.5\} = 2$; $\{5\} = 5$; $\{-2.5\} = -3$ etc.)
14. Let $\{x\}$ denote the greatest integer less than or equal to x . Some P real values of x satisfy the equation $x^2 + 10,000 \{x\} = 10,000x$. What is $P - 100$?
15. For any real number x , let $\lceil x \rceil$ denote the smallest integer that is greater than or equal to x and $\lfloor x \rfloor$ denote the largest integer that is less than or equal to x (for example, $\lceil 1.23 \rceil = 2$ and $\lfloor 1.23 \rfloor = 1$). Find the value of

$$2010 - \sum_{k=1}^{2010} \left\lceil \frac{2010}{k} \right\rceil - \left\lfloor \frac{2010}{k} \right\rfloor.$$

16. Given that

$$x = \lfloor 1^{1/3} \rfloor + \lfloor 2^{1/3} \rfloor + \lfloor 3^{1/3} \rfloor + \dots + \lfloor 7999^{1/3} \rfloor,$$

find the value of $\left\lceil \frac{x}{5000} \right\rceil$, where $\{y\}$ denotes the greatest integer less than or equal to y . (For example, $\{2.1\} = 2$, $\{30\} = 30$, $\{-10.5\} = -11$.)

17. What is the greatest integer less than or equal to $\frac{3^{100} + 2^{100}}{3^{96} + 2^{96}}$?

ANSWER KEY

EXERCISE - 2.1

Que.	1	2	3	4	5	6	7	8	9	10	11	12
Ans.	3	3	4	1	1	1	3	3	1	3	1	2

13.01	14.95	15.07	16.15	17.19	18.65	19.26
20.12	21.51	22.36	23.63	24.91	25.27	26.04
27.41	28.07	29.60	30.12	31.19	32.17	33.02
34.53	35.01	36.03				

EXERCISE - 2.2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13
Ans.	4	2	3	3	4	3	2	1	3	1	3	2	3

14.00	15.13	16.16	17.00	18.19	19.37	20.25
21.21	22.02	23.04	24.34	25.20	26.60	27.12

EXERCISE - 2.3

Que.	1	2	3	4	5	6	7	8	9	10	11	12
Ans.	3	1	4	3	1	4	2	3	2	1	3	2

13.05	14.40	15.11	16.75	17.36	18.04	19.86
20.12	21.17	22.35	23.07	24.60	25.01	26.81
27.16	28.81	29.96	30.49	31.10	32.17	33.18
34.19	35.49	36.13	37.16	38.28	39.34	40.16
41.42	42.03	43.46	44.33	45.21	46.05	47.03
48.07	49.18	50.60	51.50	52.26	53.19	54.81
55.18	56.06	57.18	58.06	59.04	60.04	61.16
62.45	63.06	64.10	65.05	66.01	67.03	68.08
69.16	70.14	71.62	72.81	73.10	74.20	75.15
76.77	77.09	78.30	79.02	80.16	81.46	82.76
83.04	84.01	85.20	86.07	87.26	88.28	89.35
90.06	91.05	92.21	93.13	94.18	95.02	96.03
97.15	98.67	99.13	100.2	101.46	102.2	103.01
104.09	105.7	106.11				

EXERCISE - 2.4

1. $\sqrt{3} + \sqrt{2}$ 2. $\sqrt{\frac{x^2-9}{x+3}}$ 3. 104 4. $a > b$ 5. $c < b < a$
6. (C), $x = 2, 3, 4, 5$ 7. -1 8. $W < V < U$ 9. $\sqrt[3]{2} + 1$ 10. 1
11. $\sqrt{5} - \sqrt{2}$ 12. $\begin{cases} 4 & \text{if } 1 \leq a < 5 \\ 2\sqrt{a-1} & \text{if } a \geq 5 \end{cases}$ 13. $4\sqrt{a(a-1)}$
14. $\frac{\sqrt{2}(\sqrt{a^2+a+1} + \sqrt{a^2-a+1})}{2}$ 15. a^2 16. 93 17. 06 18. 08
19. 10 20. 06 21. 07 22. 05 23. 32 24. 79 25. 11
26. 25 27. 04 28. 05 29. 08 30. 50

EXERCISE - 2.5

2. 9 3. 1 4. 1731 5. 88 6. 29
7. 101 8. all numbers in the form $6k + 1$ or $6k + 2$, $k = 0, 1, 2, \dots$ 10. $625 \times 10^{n-2}$ ($n \geq 2$)
11. 489 12. 1977 13. 17901 15. 512 16. (3, 2, 1), (6, 8, 4), (8, 3, 7)
17. $n = 52$ 18. 1681 19. 43 20. 20 21. 81 22. 36
23. 73 24. 08 25. 04 26. 30 27. 08

EXERCISE - 2.6

1. (4) 2. 421 3. 1972 7. $p = 3$ (unique solution) 11. 03 12. 20
13. 17 14. 09 15. 49 16. 18 17. 32 18. 04 19. 14
20. 12

EXERCISE - 2.7

2. -1 3. -2 4. 7 5. 600 6. 1486 8. $10, 7\frac{1}{2}, 9\frac{1}{8}, 8\frac{2}{7}, 6\frac{4}{5}$
9. 08 10. -9.2 11. 47 12. 50 13. 34 14. 99 15. 16
16. 23 17. 80